Before Reading Dysthe (1979)

A Prerequisite Guide to Deep-Water Wave Envelopes

3 October 2026

Abstract

This guide assumes familiarity with partial differential equations and linear algebra, but no previous course in water-wave theory or multiple-scale perturbation methods. It develops the physical and mathematical background needed to read K. B. Dysthe's 1979 paper, then works through its main stability calculation. The emphasis is on why each approximation is made, what each variable represents, and which conclusions the approximation supports.

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1 What you are preparing to understand

A nearly periodic surface wave can look regular while a small, long-scale disturbance grows on top of it. Dysthe asks how well an approximate envelope equation predicts that growth. The cubic nonlinear Schrödinger equation, abbreviated NLS, is the leading model. He retains the next order in wave steepness and finds that the current induced by a modulated wave changes the prediction substantially [1].

The paper is short because it assumes considerable background. Its hardest steps are the choices of scales, harmonic components, and approximation order that occur before the linear algebra begins. Your PDE background should make Laplace's equation, Fourier modes, and the eigenvalue calculation familiar once their physical roles are identified.

1.1 A useful reading route

Read Sections 2–5 first. They explain the wave envelope, the moving free boundary, and why a slow derivative changes the order of a term. Then study the NLS calculation in Section 6 and the mean-flow calculation in Section 7. Finally, use Sections 8–10 beside the paper. Appendix A fills in two compressed derivative calculations.

You are ready for a first reading when you can explain

1. why the carrier oscillates rapidly but its envelope changes slowly;
2. why a deep-water Fourier mode decays exponentially below the surface;
3. why group velocity is half the phase velocity;
4. why a spatially varying intensity generates a mean current;
5. why a negative radicand in the stability formula means growth.

For a derivation-level reading, also work through the determinant in Section 9 and the order reduction in Appendix A. The exercises have answers.

1.2 What is derived here and what is taken from the paper

The linear water-wave dispersion relation, NLS perturbation calculation, Fourier solution for the mean current, Dysthe stability determinant, and small-steepness growth formulas are worked out here. The numerical coefficients in the full fourth-order equation are transcribed from the scanned paper. The guide explains the harmonic-expansion procedure that produces them and the subsequent order reduction; it does not repeat every nonlinear product in the intermediate quantity Q.

Agreement with Longuet-Higgins's full-equation calculations or laboratory observations refers to *comparisons reported in the 1979 paper*. These are not new computations or independent validations. Our plots of approximate formulas do not reproduce those reference data. Perturbative derivations below are formal asymptotic calculations unless a proposition explicitly states an exact result for a specified reduced model.

1.3 Keep the different waves and frequencies separate

The *carrier* has dimensional wavenumber $k_0 > 0$ and linear angular frequency $\omega_0 = \sqrt{gk_0}$. The *modulation* is a slower variation of amplitude and phase. In the paper's dimensionless stability calculation its wavenumbers are λ, μ and its frequency is Ω. Thus Ω is not ω_0, and λ is not the carrier wavelength.

Symbol	Meaning and convention
x, y, z, t	Along-wave, transverse, vertical coordinates and time; water lies below $z = \zeta$.
g, ρ	Gravitational acceleration and constant water density.
ζ, ϕ	Real surface elevation and real velocity potential; fluid velocity is $\nabla\phi$.
a, ε	Leading crest amplitude and steepness $\varepsilon = k_0 a$; wave height is approximately $2a$.
θ	Carrier phase $k_0 x - \omega_0 t$.
A, B	Complex first-harmonic coefficients of potential and elevation. They are different variables.
$\overline{\phi}, \overline{\zeta}$	Mean potential and mean elevation; the bar denotes a carrier-period average.
$A^*, \lvert A\rvert^2$	Complex conjugate and AA^*.
A_0	Real dimensionless constant envelope amplitude; $A_0 = k_0 a$.
a', θ'	Small real amplitude and phase perturbations in (3.1); primes here are not derivatives.
λ, μ, K	Dimensionless modulation wavenumbers in x, y and $K = \sqrt{\lambda^2 + \mu^2}$.
Ω, γ	Modulation frequency and growth rate; $\gamma = \operatorname{Im}\Omega$ for the growing mode.
$P_1, P_2;\ \overline{P}_1, \overline{P}_2$	Differential operators in (3.2)–(3.3), and algebraic quantities used in the dispersion relation.

Coordinates are dimensional until Section 6.1, and thereafter use Dysthe's dimensionless convention unless stated otherwise. Paper equation numbers appear as Dysthe (2.19); other equation numbers refer to this guide. Printed pages 105–114 correspond to PDF pages 2–11 because the supplied file begins with a JSTOR cover page.

2 From a surface wave to a complex envelope

2.1 Geometry, steepness, and deep water

A linear plane wave travelling in the positive x direction has elevation

$$\zeta(x,t) = a\cos(k_0 x - \omega_0 t).$$

Its wavelength is $2\pi/k_0$, its period is $2\pi/\omega_0$, and its maximum slope has magnitude $k_0 a$. Steepness, rather than amplitude alone, measures nonlinearity: a metre-high wave can be gentle at one wavelength and steep at another.

Assumption 2.1 (Physical setting)**.** The fluid is incompressible, inviscid, and irrotational, with constant density; gravity supplies the restoring force. Surface tension, viscosity, wind forcing, and wave breaking are excluded. The depth is treated as infinite, the background is at rest, and $k_0 a \ll 1$.

At finite depth h, the carrier is in the deep-water regime when $k_0 h \gg 1$. The infinite-depth idealization is stronger: a modulation-induced mean response has depth scale $1/K_{\mathrm{dim}}$, often much larger than $1/k_0$. A bottom negligible for the carrier need not be negligible for a very long modulation. Dysthe uses an infinite lower half-space for both.

2.2 Why use a complex variable for a real water surface?

If $B = be^{i\beta}$, Euler's formula gives

$$Be^{i\theta} + B^* e^{-i\theta} = 2b\cos(\theta + \beta).$$

Magnitude and phase are stored together. A first harmonic written as $Be^{i\theta}+$c.c. has crest amplitude $2|B|$, not $|B|$. Here c.c. means to add the complex conjugate of the oscillatory terms; a real mean component is included once.

A modulated surface is represented by

$$\zeta = \overline{\zeta} + Be^{i\theta} + B^* e^{-i\theta} + \text{higher harmonics}. \tag{2.1}$$

$B(x,y,t)$ changes little over one carrier wavelength or period. Its magnitude controls local amplitude; its argument controls local phase. Modulation includes changes in crest spacing and timing as well as height.

2.3 Harmonics, sidebands, and the mean

For a real narrowband wave there is a spectral cluster near $+k_0$ and a conjugate cluster near $-k_0$. Narrowband means the width Δk of each cluster satisfies $\Delta k/k_0 \ll 1$.

A modulation creates *sidebands* close to the carrier:

$$a[1+\delta\cos(qx)]\cos(k_0 x) = a\cos(k_0 x) + \frac{a\delta}{2}[\cos((k_0+q)x) + \cos((k_0-q)x)].$$

These wavenumbers are near k_0 for $|q| \ll k_0$. A second *harmonic*, in contrast, has phase 2θ, near $2k_0$. Nonlinear products generate harmonics even for a uniform wavetrain. A *mean* component has no fast carrier oscillation.

For $f = Be^{i\theta} + B^* e^{-i\theta}$,

$$f^2 = B^2 e^{2i\theta} + 2|B|^2 + (B^*)^2 e^{-2i\theta}. \tag{2.2}$$

A carrier-period average keeps $2|B|^2$ and removes the oscillatory terms. A cubic product has first-harmonic coefficient $3|B|^2 B$, explaining the basic cubic structure of an envelope nonlinearity.

The average holds the slow coordinates effectively fixed. It is not an average over the whole packet; its result can still vary in x, y, t.

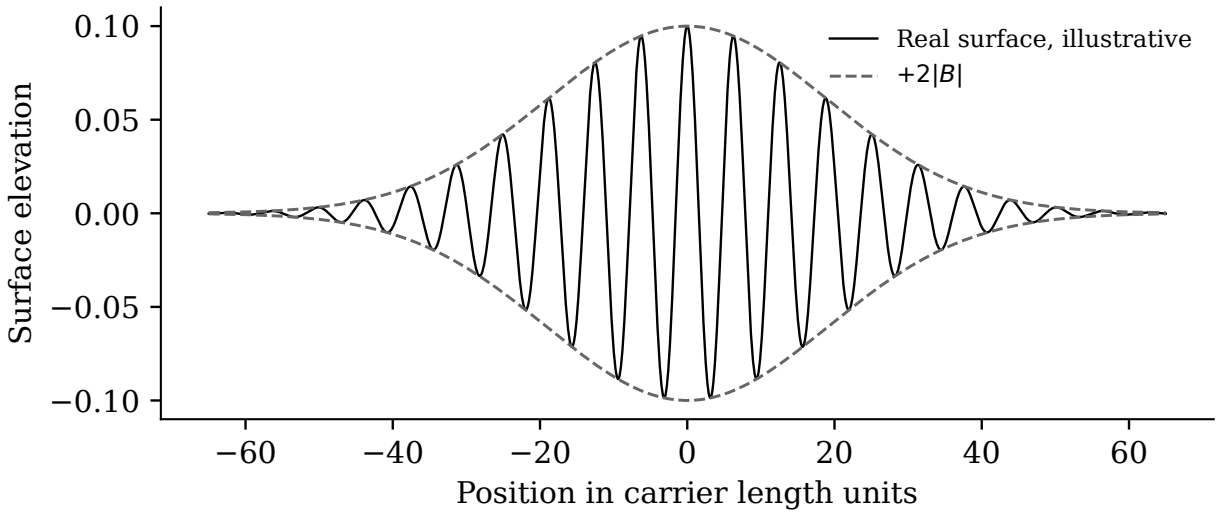


Figure 1: An illustrative carrier inside a slowly varying envelope at one instant. Dashed curves are $\pm 2|B|$. This is a schematic reconstruction, not a computed Dysthe solution.

2.4 Potential envelope versus elevation envelope

Dysthe writes the first-harmonic potential as

$$\phi = \overline{\phi} + A_{\mathrm{dim}} e^{k_0 z} e^{i\theta} + A^*_{\mathrm{dim}} e^{k_0 z} e^{-i\theta} + \cdots .$$

The exponential describes depth decay. At leading linear order, the dynamic surface condition gives

$$B_{\mathrm{dim}} = \frac{i\omega_0}{g} A_{\mathrm{dim}}. \tag{2.3}$$

For a uniform carrier, $a = 2|B_{\mathrm{dim}}|$. Hence the scaled potential envelope in Dysthe (2.18) has

$$|A| = \frac{2k_0^2}{\omega_0} |A_{\mathrm{dim}}| = k_0 a.$$

Equal leading magnitudes do not make the envelopes interchangeable at fourth order: their higher-order relations contain corrections.

Return to the paper: Read (2.6), printed p. 107. Identify the carrier, bound harmonics, and real mean fields.

Readiness check. Explain why c.c. makes the field real and why the leading surface amplitude is $2|B|$.

3 The fluid problem behind the envelope equation

3.1 Why Laplace's equation governs the fluid interior

Let $\boldsymbol{u} = (u, v, w)$ denote velocity. Incompressibility is $\nabla \cdot \boldsymbol{u} = 0$; irrotationality is $\nabla \times \boldsymbol{u} = 0$. For a smooth flow in a simply connected domain, the latter permits $\boldsymbol{u} = \nabla\phi$. Therefore

$$\nabla^2\phi = 0, \qquad z < \zeta(x, y, t). \tag{3.1}$$

This is (2.1). It is a constraint at each time, not a wave equation in the fluid interior. Time evolution enters through the surface conditions.

3.2 Kinematic condition: the surface moves with the water

Write the surface as $F = z - \zeta(x, y, t) = 0$. A surface fluid particle must remain there, so $DF/Dt = F_t + \boldsymbol{u} \cdot \nabla F = 0$. Thus

$$\zeta_t + \phi_x\zeta_x + \phi_y\zeta_y = \phi_z, \qquad z = \zeta. \tag{3.2}$$

The left side includes horizontal advection of a sloping surface. $\zeta_t = \phi_z$ alone is the linearized condition.

3.3 Dynamic condition: atmospheric pressure at the surface

Integration of the inviscid potential-flow momentum equation gives

$$\phi_t + \frac{1}{2}|\nabla\phi|^2 + \frac{p}{\rho} + gz = C(t).$$

An additive time-dependent change to ϕ absorbs $C(t)$. Subtract constant atmospheric pressure and restrict to the surface:

$$\phi_t + \frac{1}{2}|\nabla\phi|^2 + g\zeta = 0, \qquad z = \zeta. \tag{3.3}$$

This is Bernoulli's equation. Inviscid does not mean linear: $|\nabla\phi|^2$ remains.

Dysthe's (2.2) differentiates Bernoulli along a surface fluid particle. For smooth f, $Df/Dt = f_t + \nabla\phi \cdot \nabla f$. Using the particle's vertical velocity ϕ_z gives

$$\phi_{tt} + g\phi_z + \partial_t|\nabla\phi|^2 + \frac{1}{2}\nabla\phi \cdot \nabla|\nabla\phi|^2 = 0, \qquad z = \zeta. \tag{3.4}$$

Differentiation of ϕ_t contributes $\nabla\phi \cdot \nabla\phi_t$; differentiation of $\frac{1}{2}|\nabla\phi|^2$ contributes another equal term. Together they give $\partial_t|\nabla\phi|^2$, explaining its coefficient in (2.2).

3.4 Why expand the boundary condition about $z = 0$?

For smooth f and small ζ,

$$f(x, y, \zeta, t) = f(x, y, 0, t) + \zeta f_z(x, y, 0, t) + \frac{1}{2}\zeta^2 f_{zz}(x, y, 0, t) + \cdots. \tag{3.5}$$

This moves the equations to a fixed plane while retaining the effects of surface displacement as extra terms. It does not flatten the actual wave.

In (2.4), $L = \partial_t^2 + g\partial_z$. Taylor expansion of (3.4) gives

$$L\phi + \zeta L\phi_z + \partial_t|\nabla\phi|^2 + Q = 0, \qquad z = 0,$$

where the retained remainder is

$$Q = \frac{1}{2}\zeta^2 L\phi_{zz} + \frac{1}{2}\nabla\phi\cdot\nabla|\nabla\phi|^2 + \zeta\partial_{zt}|\nabla\phi|^2.$$

These pieces come from another Taylor term, the cubic material derivative, and the Taylor correction to the quadratic velocity term.

The expanded kinematic condition (2.5) is

$$\zeta_t - \phi_z + \nabla_h\cdot(\zeta\nabla_h\phi) + \nabla_h\cdot\left(\frac{1}{2}\zeta^2\nabla_h\phi_z\right) = 0, \qquad z = 0,$$

at the indicated order, where $\nabla_h = (\partial_x, \partial_y, 0)$. For example, its quadratic terms are initially $\nabla_h\phi\cdot\nabla_h\zeta - \zeta\phi_{zz}$. Since $-\phi_{zz} = \nabla_h^2\phi$, their sum becomes $\nabla_h\cdot(\zeta\nabla_h\phi)$. The divergence form is an algebraic reorganization, not an extra hypothesis.

3.5 A first nonlinear calculation: the second Stokes harmonic

A *Stokes wavetrain* is a periodic travelling solution of the nonlinear water-wave equations. Its surface is not exactly sinusoidal. Here is a short calculation showing how a bound harmonic enters.

Suppress y and let $\theta = k_0x - \omega_0t$, with $\omega_0^2 = gk_0$. The leading real potential is

$$\phi_1 = \frac{a\omega_0}{k_0}e^{k_0z}\sin\theta.$$

Through second order, try a bound elevation harmonic $\zeta = a\cos\theta + C\cos(2\theta)$ and no second potential harmonic. Expand Bernoulli at $z = 0$. Its second-order part is

$$-ga^2k_0\cos^2\theta + \frac{1}{2}a^2\omega_0^2 + gC\cos(2\theta) = 0.$$

Since $\cos^2\theta = (1+\cos 2\theta)/2$, this gives

$$C = \frac{1}{2}k_0a^2, \qquad \zeta = a\cos\theta + \frac{1}{2}k_0a^2\cos(2\theta) + \cdots. \tag{3.6}$$

The kinematic condition independently gives $2\omega_0C = k_0\omega_0a^2$, the same result. At a crest both elevation terms are positive, while at a trough the second term is still positive; the profile has sharper crests and flatter troughs. The harmonic oscillates at $2\omega_0$, whereas a free linear mode of wavenumber $2k_0$ has frequency $\sqrt{2}\omega_0$. It is therefore forced by the carrier, not an independent linear wave. This simple calculation illustrates the Stokes comparison mentioned immediately after (2.13).

Return to the paper: Read (2.1)–(2.5), printed pp. 106–107. Connect each equation to incompressibility, surface motion, or pressure.

Readiness check. Derive (3.2) from $DF/Dt = 0$ and identify which terms disappear upon linearization.

4 Linear deep-water waves and dispersion

4.1 Deriving the carrier dispersion relation

Linearize about a flat, resting surface and evaluate at $z = 0$:

$$\zeta_t = \phi_z, \qquad \phi_t = -g\zeta. \tag{4.1}$$

Take one complex Fourier component

$$\phi = Ce^{k_0 z}e^{i(k_0 x - \omega_0 t)}, \qquad \zeta = De^{i(k_0 x - \omega_0 t)}.$$

Real fields include the conjugates. Laplace's equation holds because ∂_x^2 contributes $-k_0^2$ and ∂_z^2 contributes $+k_0^2$. $e^{k_0 z}$ decays as $z \to -\infty$; the growing exponential is excluded.

The surface conditions give

$$-i\omega_0 D = k_0 C, \qquad -i\omega_0 C = -gD.$$

Elimination yields

$$\boxed{\omega_0^2 = gk_0.} \tag{4.2}$$

This also verifies (2.3). For two horizontal directions, replace k_0 by the horizontal wavevector's magnitude.

4.2 Phase velocity and group velocity

A crest has fixed phase $k_0 x - \omega_0 t$, so $c_p = \omega_0/k_0$. To find packet velocity, add two nearby waves:

$$\begin{aligned}&\cos((k_0+q)x - (\omega_0+\delta\omega)t) + \cos((k_0-q)x - (\omega_0-\delta\omega)t)\\ &\quad = 2\cos(qx - \delta\omega t)\cos(k_0 x - \omega_0 t).\end{aligned}$$

The slow factor travels at $\delta\omega/q$, tending to $c_g = \omega'(k_0)$ in the narrowband limit. Thus

$$c_g = \frac{\omega_0}{2k_0} = \frac{1}{2}c_p. \tag{4.3}$$

Crests move through the group. The group is neither a particular crest nor a collection of the same fluid particles.

$X = x - c_g t$ follows leading packet translation. $\partial_t + c_g\partial_x$ measures evolution in that moving frame; it is the transport derivative in (2.8).

4.3 Dispersion curvature gives relative phase evolution

For a nearby longitudinal wavenumber,

$$\omega(k_0+q) = \omega_0 + c_g q + \tfrac{1}{2}\omega''(k_0)q^2 + \cdots, \qquad \omega''(k_0) = -\frac{\omega_0}{4k_0^2}.$$

The constant is carrier frequency, the linear term gives translation, and the quadratic term controls relative phase evolution within the packet. In a PDE it is represented by a second derivative of the envelope.

Transverse curvature has the opposite sign. With dimensionless longitudinal offset q and transverse offset r,

$$\frac{\omega(k_0(1+q), k_0 r)}{\omega_0} = 1 + \frac{q}{2} - \frac{q^2}{8} + \frac{r^2}{4} + \frac{q^3}{16} - \frac{3qr^2}{8} + O((|q|+|r|)^4). \tag{4.4}$$

This explains the quadratic terms in NLS and the third derivatives in Dysthe's equation. To check a PDE coefficient, insert $e^{i(qx+ry-\nu t)}$ and compare its ν with this polynomial.

Example 4.1 (A scale check). For wavelength 10 m, $k_0 = 2\pi/10\,\mathrm{m}^{-1}$. Taking $g = 9.81\,\mathrm{m\,s}^{-2}$ gives $\omega_0 \approx 2.48\,\mathrm{s}^{-1}$, $c_p \approx 3.95\,\mathrm{m\,s}^{-1}$, and $c_g \approx 1.98\,\mathrm{m\,s}^{-1}$. For $a = 0.10\,\mathrm{m}$ the steepness is approximately 0.063. These are illustrative linear values, not breaking-wave predictions.

4.4 What the deep-water limit removes

For a flat bottom at $z = -h$, the no-normal-flow condition is $\phi_z(-h) = 0$. A linear horizontal mode then has vertical factor $\cosh[k_0(z+h)]$ rather than $e^{k_0 z}$. At the surface its logarithmic vertical derivative is

$$\frac{\phi_z}{\phi} = k_0 \tanh(k_0 h).$$

The same two linear surface conditions therefore give

$$\omega_0^2 = g k_0 \tanh(k_0 h).$$

For $k_0 h \gg 1$, $\tanh(k_0 h)$ tends to one and (4.2) is recovered. For $k_0 h \ll 1$, $\tanh(k_0 h) \approx k_0 h$ and $\omega_0 \approx k_0\sqrt{gh}$. The shallow-water limiting phase and group speeds are equal; the deep-water factor of two is a consequence of a different dispersion law.

This derivation explains why depth is a model assumption, not merely a descriptive adjective. A change of depth modifies dispersion and the mean-flow boundary problem, so one cannot retain the deep-water envelope coefficients unchanged. Dysthe mentions finite-depth extensions in the introduction, but the equations derived in this paper use infinite depth.

Return to the paper: Read (2.7)–(2.8), p. 107, and the dispersive terms of (2.19), p. 109. Optional reinforcement: [2].

Readiness check. Recover the q^2 and r^2 signs in (4.4) from the expansion of $(1+s)^{1/4}$.

5 Weak nonlinearity and multiple-scale bookkeeping

5.1 Two small quantities, with a chosen relationship

Weak nonlinearity means $\varepsilon = k_0 a \ll 1$. Slow modulation means $\delta = \Delta k/k_0 \ll 1$. They are distinct requirements. Dysthe balances them by taking $\delta = O(\varepsilon)$. A small amplitude alone does not make an arbitrary spectrum narrowband.

After choosing carrier units, display the scales as

$$X = \varepsilon(x - c_g t), \qquad Y = \varepsilon y, \qquad T = \varepsilon^2 t.$$

If $A = \varepsilon\alpha(X, Y, T)$, the chain rule gives

$$\begin{aligned} A_x &= \varepsilon^2 \alpha_X, & A_{xx} &= \varepsilon^3 \alpha_{XX}, \\ A_t &= -c_g \varepsilon^2 \alpha_X + \varepsilon^3 \alpha_T, & (\partial_t + c_g \partial_x) A &= \varepsilon^3 \alpha_T. \end{aligned}$$

The transport cancellation is essential: A_t alone can be second order, while the derivative following the group is third order.

Definition 5.1 (Formal order)**.** Here $f = O(\varepsilon^m)$ means that the scaled coefficient of ε^m is treated as bounded in the chosen slow variables. Products add orders. A derivative acting on a slow profile adds a factor of ε, apart from the slower group-frame time scale.

A rigorous bound would specify the domain, norm, derivative bounds, and time interval, and prove uniform estimates. The paper's order statements are formal bookkeeping, not an error theorem for arbitrary initial data.

5.2 Distinguish slow derivatives from fast derivatives

For the full oscillatory field,

$$\partial_x (A e^{i\theta}) = (A_x + i k_0 A) e^{i\theta}.$$

The carrier derivative $ik_0 A$ does not raise the slow order. The statement after (2.19) that a derivative adds an order concerns derivatives of the slow coefficients. Carrier derivatives have already been extracted.

The vertical mean-flow scale is slow as well. Its depth dependence is e^{Kz} with $K = O(\varepsilon)$ in carrier units, whereas the carrier has e^z dependence.

Quantity in carrier units	Formal order
A	ε
A_x, A_y	ε^2
$\mathcal{D}A$, $\mathcal{D} = \partial_t + \frac{1}{2}\partial_x$	ε^3
$A_{xx}, A_{yy}, A\lvert A\rvert^2$	ε^3
$A_{xxx}, A_{xyy}, \lvert A\rvert^2 A_x, A^2 A_x^*$	ε^4
$\overline{\phi}$	ε^2
$\overline{\phi}_x, \overline{\phi}_z, \overline{\zeta}$	ε^3
$A\overline{\phi}_x, A\overline{\phi}_z$	ε^4

NLS keeps the third-order balance; Dysthe also keeps fourth-order terms. "Fourth order" refers to the perturbation expansion, not a fourth spatial derivative or a quartic nonlinearity.

5.3 Coefficient matching and resonance

Insert the harmonic expansion into the boundary conditions. Collect the coefficient of each $e^{im\theta}$ and the slow mean $m = 0$ separately. Higher harmonics are often determined by forced linear problems in terms of A. They are *bound*: their phase follows the carrier rather than an independent linear frequency.

The fundamental $m = 1$ is special. The leading linear operator already has the carrier as a homogeneous solution; forcing at that phase can resonate. The requirement that the next correction remain bounded supplies the envelope evolution equation.

For comparison, $u_{tt} + u = \varepsilon \cos t$ has a particular term $\frac{1}{2}\varepsilon t \sin t$, which grows at long times. Allowing the leading amplitude and phase to evolve slowly removes this secular forcing from the next correction. This principle produces NLS and its next correction.

The cubic product in Section 2 explains the structure $|A|^2 A$ but not its water-wave coefficient. That coefficient comes from the full boundary conditions, lower harmonics, and normalization.

5.4 Order reduction

An intermediate equation contains A_{tt}. Leading transport gives $A_t \approx -\frac{1}{2}A_x$, hence $A_{tt} \approx \frac{1}{4}A_{xx}$. For fourth-order accuracy one must also differentiate the third-order NLS correction. Appendix A.2 shows this calculation. Such substitutions are justified only when their errors lie beyond the target order; they are not identities for arbitrary functions.

Return to the paper: Read the order statements around (2.6), (2.12)–(2.17), and after (2.19). Identify which derivatives act on slow coefficients.

Readiness check. For $A = \varepsilon\alpha(\varepsilon x)$, check that $|A|^2 A_x$ is fourth order and $|A|^2 A$ third order.

6 NLS: the baseline approximation and its instability

6.1 Dysthe's dimensionless variables

From this point, use the transformation in (2.18). To make its meaning unambiguous, write

$$t = \omega_0 t_{\rm dim}, \qquad (x,y,z) = k_0(x_{\rm dim}, y_{\rm dim}, z_{\rm dim}),$$
$$A = \frac{2k_0^2}{\omega_0} A_{\rm dim}, \qquad \overline{\phi} = \frac{2k_0^2}{\omega_0}\overline{\phi}_{\rm dim}, \qquad \overline{\zeta} = k_0 \overline{\zeta}_{\rm dim}.$$

The carrier phase becomes $x - t$ and $c_g = 1/2$. These are carrier units, not yet the separate slow coordinates X, Y, T. The envelope remains $O(\varepsilon)$ and varies slowly in x, y, t.

Deleting the right side of (2.19) gives

$$2i\left(A_t + \frac{1}{2}A_x\right) + \frac{1}{2}A_{yy} - \frac{1}{4}A_{xx} - |A|^2 A = 0. \tag{6.1}$$

In the group coordinate $X = x - t/2$, a longitudinal envelope obeys

$$iA_t - \frac{1}{8}A_{XX} - \frac{1}{2}|A|^2 A = 0.$$

Both coefficients are negative. Their product is positive, so this has the focusing NLS sign combination; reversing time gives the frequently used convention with two positive signs. A sign difference between references need not imply different physics, but conventions must be checked.

The dispersive coefficients follow from (4.4). The cubic term describes an amplitude-dependent frequency shift. A constant envelope solution is

$$A = A_0 e^{-iA_0^2 t/2}, \qquad A_0 > 0. \tag{6.2}$$

Its magnitude is constant. Multiplication by the carrier gives total phase $x - (1 + A_0^2/2)t$ to this order. The extra phase rotation is the weakly nonlinear Stokes frequency correction.

6.2 What linear stability asks

Stability asks whether a small disturbance of this particular solution remains small initially or grows. It does not ask whether $|A|$ itself oscillates, and it does not prove later breaking.

Introduce an independent small perturbation parameter η:

$$A = A_0[1 + \eta u(x,y,t)]e^{i(\eta v(x,y,t) - A_0^2 t/2)}, \tag{6.3}$$

where u, v are real and $\eta \ll 1$. To first order, $A = A_0 e^{-iA_0^2 t/2}[1 + \eta(u + iv)] + O(\eta^2)$. The amplitude-expansion parameter ε and perturbation parameter η have different jobs. Linearization discards η^2 while retaining the prescribed powers of $A_0 = O(\varepsilon)$.

Set $\mathcal{D} = \partial_t + \frac{1}{2}\partial_x$ and $P_2 = \frac{1}{4}\partial_y^2 - \frac{1}{8}\partial_x^2$. Substituting (6.3) into (6.1), cancelling the background solution, and separating real and imaginary parts gives

$$\mathcal{D}u + P_2 v = 0, \qquad \mathcal{D}v - P_2 u + A_0^2 u = 0. \tag{6.4}$$

One useful intermediate identity is $|A|^2 A = A_0^3 e^{-iA_0^2 t/2}[1 + \eta(3u + iv)] + O(\eta^2)$. The phase shift from the background cancels one part of this contribution.

6.3 The Fourier-mode matrix calculation

Use a complex representation for the perturbation modes and take real parts at the end:

$$(u, v) = (\widehat{u}, \widehat{v})e^{i(\lambda x+\mu y-\Omega t)}.$$

Define

$$p = \frac{\lambda^2}{8} - \frac{\mu^2}{4}, \qquad V = \Omega - \frac{\lambda}{2}.$$

$\mathcal{D}$ acts as $-iV$ and P_2 acts as p. The homogeneous system is

$$\begin{pmatrix} -iV & p \\ A_0^2 - p & -iV \end{pmatrix} \begin{pmatrix} \widehat{u} \\ \widehat{v} \end{pmatrix} = 0.$$

A nonzero perturbation requires determinant zero:

$$\left(\Omega - \frac{\lambda}{2}\right)^2 = p(p - A_0^2). \tag{6.5}$$

Proposition 6.1 (Growth in the linearized NLS model)**.** *For $A_0 > 0$ and real λ, μ, a growing Fourier mode exists precisely when $0 < p < A_0^2$. Its growth rate is $\gamma = \sqrt{p(A_0^2 - p)}$.*

Proof. If $0 < p < A_0^2$, the right side of (6.5) is negative, so one frequency has $\operatorname{Im}\Omega > 0$. For $\Omega = \Omega_r + i\gamma$, the time factor is $e^{-i\Omega t} = e^{-i\Omega_r t}e^{\gamma t}$. Otherwise the roots are real, including repeated roots at the boundaries, and have no positive exponential growth rate. □

Repeated boundary roots can require separate analysis of neutral or algebraic behavior. The proposition concerns exponential mode growth only.

For longitudinal perturbations, $\mu = 0$ and $p = \lambda^2/8$. Thus

$$0 < |\lambda| < 2\sqrt{2}A_0, \qquad |\lambda_M| = 2A_0, \qquad \gamma_M = \frac{1}{2}A_0^2. \tag{6.6}$$

To maximize growth, complete the square: $p(A_0^2 - p) = A_0^4/4 - (p - A_0^2/2)^2$. The fastest NLS disturbance therefore has $p = A_0^2/2$.

6.4 Why transverse perturbations matter

For two horizontal modulation directions, the maximum condition is

$$\lambda^2 - 2\mu^2 = 4A_0^2. \tag{6.7}$$

This hyperbola is unbounded. The NLS polynomial dispersion approximation formally predicts equal maximum growth at arbitrarily large modulation wavenumbers along it, although those wavenumbers violate the narrowband assumption. Dysthe's correction reduces this degeneracy. It does not make a small-bandwidth approximation valid at large bandwidth.

Return to the paper: Read the introduction's discussion of NLS, then (3.1)–(3.7). The matrix calculation above is the simpler version of Dysthe's calculation.

Readiness check. Explain why Ω with positive imaginary part grows for the convention $e^{-i\Omega t}$.

7 The induced mean current: Dysthe's central new effect

7.1 Averaging reveals a response invisible in the carrier alone

Quadratic wave quantities such as velocity products have a nonzero carrier-period average. The wave therefore carries averaged momentum flux. The wave contribution to that flux is called *radiation stress*. Here radiation means transport by waves, not electromagnetic radiation. A uniform flux has no horizontal divergence; a spatially modulated flux can exert an averaged force and generate a mean-flow response.

This is a physical interpretation. Dysthe determines the mean response from the averaged potential-flow surface conditions, not by introducing a separate viscous forcing law. The induced current remains irrotational under the paper's hypotheses.

In dimensional variables his (2.9)–(2.11) read

$$\begin{aligned} \overline{\phi}_t + g\overline{\zeta} &= 0, && z = 0, \\ \overline{\phi}_z - \overline{\zeta}_t &= \frac{2\omega_0}{g}\boldsymbol{k}_0 \cdot \nabla_h |A_{\mathrm{dim}}|^2, && z = 0, \\ \nabla^2\overline{\phi} &= 0, && z < 0. \end{aligned}$$

The dimensions of the forcing and of $\overline{\phi}_z$ are both velocity. In carrier-scaled variables, these become

$$\overline{\phi}_t + 2\overline{\zeta} = 0, \qquad \overline{\phi}_z - 2\overline{\zeta}_t = \partial_x |A|^2, \qquad \nabla^2\overline{\phi} = 0. \tag{7.1}$$

The first equation gives $\overline{\zeta} = O(\varepsilon^3)$ because $\overline{\phi} = O(\varepsilon^2)$ and its time derivative is slow. Thus $\overline{\zeta}_t = O(\varepsilon^4)$, whereas $\overline{\phi}_z$ and $\partial_x |A|^2$ are third order. At the retained order,

$$\overline{\phi}_z = \partial_x |A|^2 \quad (z = 0), \qquad \nabla^2\overline{\phi} = 0 \quad (z < 0). \tag{7.2}$$

Together with decay at depth and a choice of zero background current, this determines the mean velocity. An additive potential constant is a gauge freedom and has no physical velocity.

7.2 Solve the boundary problem one Fourier mode at a time

Let $I = |A|^2$, and suppose it has a Fourier component $\widehat{I} e^{i(\lambda x + \mu y)}$ with $K = \sqrt{\lambda^2 + \mu^2} > 0$. Seeking a decaying solution gives

$$\overline{\phi} = C e^{Kz} e^{i(\lambda x + \mu y)}.$$

The surface condition is $KC = i\lambda \widehat{I}$, so

$$\widehat{\overline{\phi}}(0) = \frac{i\lambda}{K}\widehat{I}, \qquad \boxed{\widehat{\overline{\phi}_x}(0) = -\frac{\lambda^2}{K}\widehat{I}.} \tag{7.3}$$

The horizontal current has a negative real Fourier multiplier. This is exactly the source of the λ^2/K term in the stability result.

At $K = 0$, the forcing is zero. Set the derivative response to zero; do not divide by K. A uniform wavetrain generates no mean-current *forcing in this model*. This does not mean that water under a uniform Stokes wave has no particle drift. Eulerian induced mean flow and Lagrangian Stokes drift are different quantities.

7.3 Nonlocality and the Hilbert transform

With no y dependence, $K = |\lambda|$ and the current multiplier is $-|\lambda|$. Define $|D_x|$ by the Fourier multiplier $|\lambda|$. For the transform convention

$$\widehat{f}(\lambda) = \int_{\mathbb{R}} f(x) e^{-i\lambda x}\, \mathrm{d}x,$$

the Hilbert transform H has multiplier $-i \operatorname{sgn} \lambda$, so $H\partial_x = |D_x|$. Consequently

$$\overline{\phi}_x(x,0) = -|D_x| I(x) = -H\partial_x I(x). \tag{7.4}$$

This is nonlocal: knowing intensity at one point is not sufficient to calculate the current there. One must solve the half-space problem, or apply its Fourier multiplier to the full profile. Periodic profiles use the corresponding discrete multiplier and zero mode.

Example 7.1 (An explicitly solvable intensity modulation). Let $I(x) = I_0 + I_1 \cos(\kappa x)$, where $\kappa > 0$ and $|I_1| < I_0$. Then

$$\overline{\phi} = -I_1 e^{\kappa z} \sin(\kappa x), \qquad \overline{\phi}_x(x,0) = -\kappa I_1 \cos(\kappa x).$$

Differentiate to verify both Laplace's equation and $\overline{\phi}_z = I_x$ at $z = 0$. For $I_1 > 0$, the current response at an intensity maximum is opposite the carrier propagation direction under this convention. The response penetrates depth $1/\kappa$.

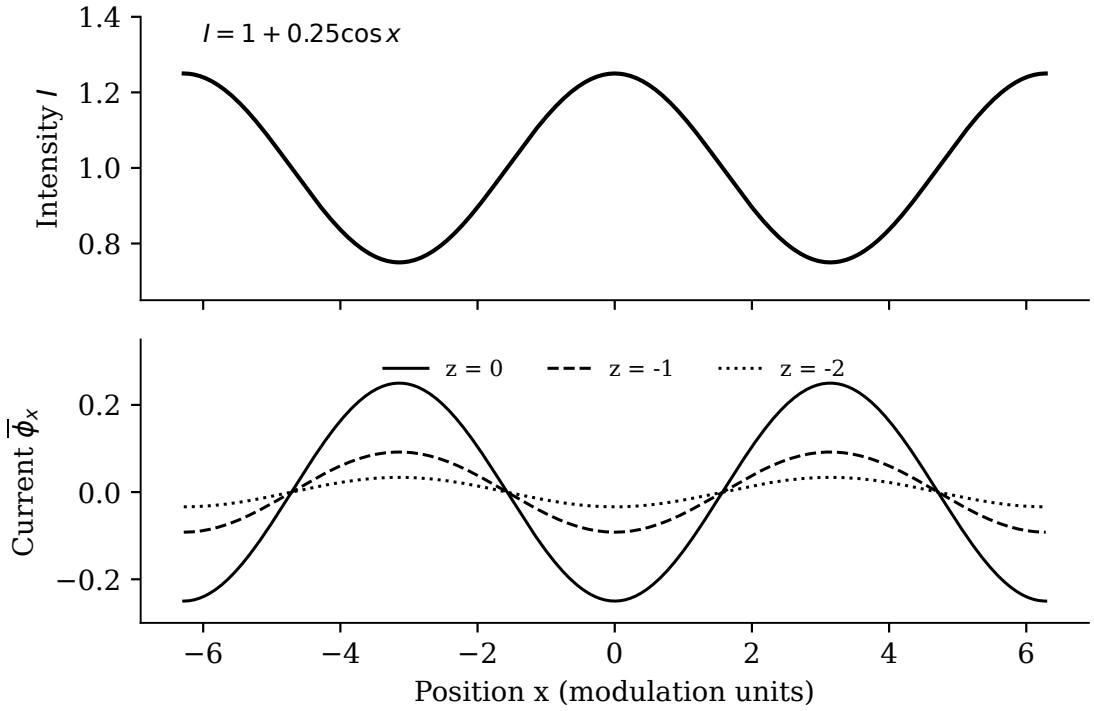


Figure 2: The cosine intensity example. The mean horizontal current $\overline{\phi}_x$ is in opposite phase to the intensity variation, and its amplitude decays as $e^{\kappa z}$. Vertical profiles use illustrative values; the curves depict the analytic half-space solution.

7.4 Why a small current affects wave stability

For a weak, slowly varying along-wave current U, the local dimensional frequency has a Doppler contribution k_0U. In Dysthe's scaled envelope equation the current is represented by $A\overline{\phi}_x$. Because the physical scaling gives $k_0U/\omega_0 = \overline{\phi}_x/2$, a locally uniform current produces the expected envelope phase shift after the factor $2i$ is accounted for.

The intensity changes the current, the current changes local phase, and phase changes feed back on the modulation. It is a coupled stability mechanism. Calling it detuning does not imply that every disturbance is stabilized or that the wave loses energy to viscosity.

Return to the paper: Read (2.9)–(2.11), (2.20)–(2.21), and (4.1)–(4.3). Then read the four physical steps on printed p. 114.

Readiness check. Derive the multiplier $-\lambda^2/K$ without using a Hilbert-transform identity.

8 Reading the fourth-order envelope equation

8.1 The equation in the paper's convention

At $z = 0$, Dysthe (2.19) is

$$2i\left(A_t + \frac{1}{2}A_x\right) + \frac{1}{2}A_{yy} - \frac{1}{4}A_{xx} - A|A|^2 = -\frac{i}{8}(6A_{xyy} - A_{xxx}) + \frac{3i}{2}A(AA_x^* - A^*A_x) + \frac{i}{2}|A|^2A_x + A(\overline{\phi}_x - i\overline{\phi}_z). \tag{8.1}$$

Subscripts denote partial derivatives and $A_x^* = \partial_x(A^*)$. The mean field obeys (7.1), or its consistently reduced version (7.2). The right side is fourth order under Section 5's ordering. The left side is the NLS balance.

Term	Role
$2i(A_t + A_x/2)$	Envelope change following the group.
$A_{yy}/2 - A_{xx}/4$	Leading transverse and longitudinal dispersion.
$-A\|A\|^2$	Leading nonlinear phase shift.
$-i(6A_{xyy} - A_{xxx})/8$	Next Taylor terms of linear dispersion.
$\frac{3i}{2}A(AA_x^* - A^*A_x) + \frac{i}{2}\|A\|^2A_x$	Nonlinear effects involving envelope gradients.
$A(\overline{\phi}_x - i\overline{\phi}_z)$	Coupling to the induced mean response.

The combination $AA_x^* - A^*A_x$ is imaginary. If $A = Re^{i\psi}$ with $R > 0$, it equals $-2iR^2\psi_x$. Its role therefore involves a local phase gradient. The term $\partial_x|A|^2$ instead measures amplitude variation. These structures need not have the same effect even though both are slow.

8.2 An algebraic evolution form of the printed equation

Divide (8.1) by $2i$, move the right side to the left, and use $\overline{\phi}_z = (|A|^2)_x$. At the retained order this gives

$$A_t + \frac{1}{2}A_x + \frac{i}{8}A_{xx} - \frac{i}{4}A_{yy} + \frac{i}{2}|A|^2A - \frac{1}{16}A_{xxx} + \frac{3}{8}A_{xyy} + |A|^2A_x - \frac{1}{4}A^2A_x^* + \frac{i}{2}A\overline{\phi}_x = 0. \tag{8.2}$$

The coefficient change in the gradient terms uses the vertical mean-flow boundary condition. For example, $\frac{1}{2}A\overline{\phi}_z = \frac{1}{2}|A|^2A_x + \frac{1}{2}A^2A_x^*$. This is why comparing partially reduced equations can make coefficients appear inconsistent. Always record which mean-field terms have been eliminated.

For a longitudinal profile, (7.4) turns the last term into $-\frac{i}{2}A|D_x|(|A|^2)$ in this temporal potential-envelope convention. Other variables and spatial evolution conventions can have different signs and coefficients. The original paper and the later spatial free-surface model are compared separately in Appendix B.

8.3 A source-consistency issue to recognize

There is an apparent coefficient inconsistency in the supplied scan. Taken literally, the printed plus sign in front of $i|A|^2A_x/2$ in (2.19) gives the evolution form (8.2). Linearizing it gives

$$P_1u + \frac{3}{4}A_0^2u_x + P_2v = 0, \qquad P_1v + \frac{5}{4}A_0^2v_x - P_2u + A_0^2u = -\frac{1}{2}\overline{\phi}_x.$$

In contrast, the paper's printed (3.2), after using $\overline{\phi}_z = 2A_0^2 u_x$, has $5A_0^2 u_x/4$; its (3.3) has $7A_0^2 v_x/4$. Changing the displayed $+i|A|^2 A_x/2$ to a minus would produce those two coefficients. This is an algebraic consistency observation about the supplied scan; no publisher erratum has been verified. The literal intermediate (2.17) also does not reproduce all the gradient coefficients of (2.19); see Appendix A.2.

For learning the main stability argument, the next section starts explicitly from the printed linearized system (3.2)–(3.3), which does reproduce (3.6)–(3.13). We preserve (2.19) as printed rather than silently editing it. You should not conclude that a failure to match these particular coefficients is a gap in your prerequisite knowledge. Nor should this guide be used as an unqualified numerical implementation specification.

8.4 What the simplified system in the conclusion means

Dysthe (4.1)–(4.3) keeps just NLS and the horizontal current:

$$2i\left(A_t + \frac{1}{2}A_x\right) + \frac{1}{2}A_{yy} - \frac{1}{4}A_{xx} = A(|A|^2 + \overline{\phi}_x), \tag{8.3}$$

with (7.2). It captures the leading improvement in the *growth-rate and instability-band* calculations emphasized in the paper. It is not an equality between complete fourth-order dynamics.

In particular, the full equation also shifts the real perturbation frequency by terms containing λA_0^2 and third powers of modulation wavenumber. Those terms matter for phase or transport even when they do not change the growth rate at the retained order. Calling the mean current dominant is a statement about the particular stability observables being studied, not about every possible wave-packet calculation.

Return to the paper: Read (2.19), p. 109, using the table above. Compare it with the explicitly simplified system on p. 113.

Readiness check. Derive (8.2) from the printed (8.1). Then compare its linearization with the source-consistency note, rather than assuming every printed line agrees.

9 Working through Dysthe's stability calculation

9.1 Linearization about a uniform nonlinear wavetrain

The uniform solution remains (6.2), with zero induced mean current and elevation. Use (6.3); Dysthe's a', θ' are our $\eta u, \eta v$. In this section the common factor η is suppressed, and the mean perturbation is first order in η.

Starting from the paper's printed equations (3.2)–(3.3), independently of the discrepancy noted in Section 8.3, gives

$$P_1 u + \frac{1}{4} A_0^2 u_x + P_2 v = -\frac{1}{2}\overline{\phi}_z, \tag{9.1}$$

$$P_1 v + \frac{7}{4} A_0^2 v_x - P_2 u + A_0^2 u = -\frac{1}{2}\overline{\phi}_x, \tag{9.2}$$

where

$$P_1 = \partial_t + \frac{1}{2}\partial_x + \frac{1}{16}(6\partial_x\partial_y^2 - \partial_x^3), \qquad P_2 = \frac{1}{4}\partial_y^2 - \frac{1}{8}\partial_x^2.$$

The mean boundary condition linearizes as

$$\overline{\phi}_z = 2A_0^2 u_x, \tag{9.3}$$

because $|A|^2 = A_0^2(1 + 2\eta u) + O(\eta^2)$.

The extra u_x term in (9.1) changes from $1/4$ to $5/4$ when its right side is moved left using (9.3).

9.2 Eliminate the mean field using the half-space solution

For the mode $e^{i(\lambda x + \mu y - \Omega t)}$ with $K > 0$,

$$\widehat{\overline{\phi}}(z) = \frac{2i\lambda A_0^2}{K}\widehat{u}\, e^{Kz}, \qquad \widehat{\overline{\phi}_x}(0) = -\frac{2\lambda^2 A_0^2}{K}\widehat{u}.$$

Define the quantities used in the paper:

$$\overline{P}_1 = \Omega - \frac{\lambda}{2} + \frac{\lambda}{16}(6\mu^2 - \lambda^2), \qquad \overline{P}_2 = \frac{\lambda^2}{8} - \frac{\mu^2}{4}. \tag{9.4}$$

P_1 acts as $-i\overline{P}_1$, while P_2 acts as $\overline{P}_2$. The bar in these two symbols is a paper convention for the algebraic quantities, not a time average of the operators.

The two perturbation equations become

$$\begin{pmatrix} -i(\overline{P}_1 - \frac{5}{4}\lambda A_0^2) & \overline{P}_2 \\ A_0^2(1 - \lambda^2/K) - \overline{P}_2 & -i(\overline{P}_1 - \frac{7}{4}\lambda A_0^2) \end{pmatrix} \begin{pmatrix} \widehat{u} \\ \widehat{v} \end{pmatrix} = 0. \tag{9.5}$$

Taking the determinant recovers (3.6):

$$(\overline{P}_1 - \tfrac{5}{4}\lambda A_0^2)(\overline{P}_1 - \tfrac{7}{4}\lambda A_0^2) = \overline{P}_2\{\overline{P}_2 - A_0^2(1 - \lambda^2/K)\}. \tag{9.6}$$

The mean current modifies the effective nonlinear factor by $1 - \lambda^2/K$. Its dependence on modulation wavenumber is essential.

9.3 Why the paper discards a term after solving the quadratic

Completing the square gives the exact roots of this truncated determinant:

$$\overline{P}_1 = \frac{3}{2}\lambda A_0^2 \pm \sqrt{\overline{P}_2\{\overline{P}_2 - A_0^2(1-\lambda^2/K)\} + \frac{1}{16}\lambda^2 A_0^4}. \tag{9.7}$$

But $A_0, \lambda, \mu = O(\varepsilon)$, so $\overline{P}_2 = O(\varepsilon^2)$. The radicand contains a leading fourth-order term and a fifth-order mean-current correction. The added $\lambda^2 A_0^4/16$ is sixth order. Other contributions of that order could come from fifth-order terms omitted from the original envelope equation. Retaining this one term would not supply a consistently more accurate physical prediction.

Dysthe therefore uses the asymptotically consistent root (3.7):

$$\overline{P}_1 \simeq \frac{3}{2}\lambda A_0^2 \pm \sqrt{\overline{P}_2\{\overline{P}_2 - A_0^2(1-\lambda^2/K)\}}. \tag{9.8}$$

Here $\simeq$ explicitly indicates truncation. Away from marginal roots, the discarded sixth-order radicand term changes the frequency at fourth order, beyond the third-order frequency correction supported by the fourth-order field equation. Near a zero radicand, such square-root expansions need extra care; do not claim a uniform error estimate there.

The real offset in (9.4) contains the higher linear dispersion; $3\lambda A_0^2/2$ is also real. They shift mode frequency but do not alter its imaginary part at this order. The growth-rate change is traced to the horizontal induced current.

9.4 Longitudinal instability band

For $\mu = 0$, $K = |\lambda|$, so the growing root has

$$\gamma^2 = \frac{\lambda^2}{8}\left[A_0^2(1-|\lambda|) - \frac{\lambda^2}{8}\right]. \tag{9.9}$$

Within the narrowband regime, the exponential instability condition is

$$0 < \lambda^2 < 8A_0^2(1-|\lambda|). \tag{9.10}$$

The explicit exclusion of $\lambda = 0$ means that the uniform perturbation has zero exponential growth. It is a neutral symmetry mode, not a growing sideband.

For $\lambda > 0$, solving the approximate marginal condition gives

$$\lambda_c = -4A_0^2 + \sqrt{8A_0^2 + 16A_0^4} = 2\sqrt{2}A_0 - 4A_0^2 + O(A_0^3).$$

This closed form is exact for the displayed approximate band equation, not an all-orders water-wave stability boundary.

The real frequency from the full retained root is

$$\operatorname{Re}\Omega = \frac{\lambda}{2} + \frac{\lambda^3}{16} + \frac{3}{2}\lambda A_0^2 \qquad (\mu = 0,\ \gamma > 0).$$

At the marginal mode, retaining through A_0^2 gives

$$\Omega_c \simeq \sqrt{2}A_0(1-\sqrt{2}A_0), \tag{9.11}$$

which is Dysthe (3.9). Higher-order frequency terms require a correspondingly higher-order determination of the cutoff.

9.5 Find the fastest growing mode

Set $s = A_0 > 0$ and use $\lambda > 0$. Differentiate (9.9):

$$\gamma^2 = \frac{s^2\lambda^2}{8} - \frac{s^2\lambda^3}{8} - \frac{\lambda^4}{64}, \qquad \frac{\mathrm{d}\gamma^2}{\mathrm{d}\lambda} = \frac{\lambda}{16}(4s^2 - 6s^2\lambda - \lambda^2).$$

The positive stationary root is $\lambda_M = -3s^2 + \sqrt{4s^2 + 9s^4}$. Expand at small s:

$$\lambda_M = 2s - 3s^2 + O(s^3). \tag{9.12}$$

Substitution into the growth formula gives $\gamma_M^2 = s^4/4 - s^5 + O(s^6)$ and hence

$$\gamma_M = \frac{1}{2}s^2(1 - 2s) + O(s^4). \tag{9.13}$$

Finally, substituting (9.12) into the real frequency and retaining through s^2 gives

$$\mathrm{Re}\,\Omega_M = s(1 - \tfrac{3}{2}s) + O(s^3). \tag{9.14}$$

These are (3.10)–(3.12). The mode and maximum-growth corrections have different absolute orders because the leading quantities have different orders. The approximate formula at $s = 0.5$ must not be interpreted as an established complete stabilization threshold.

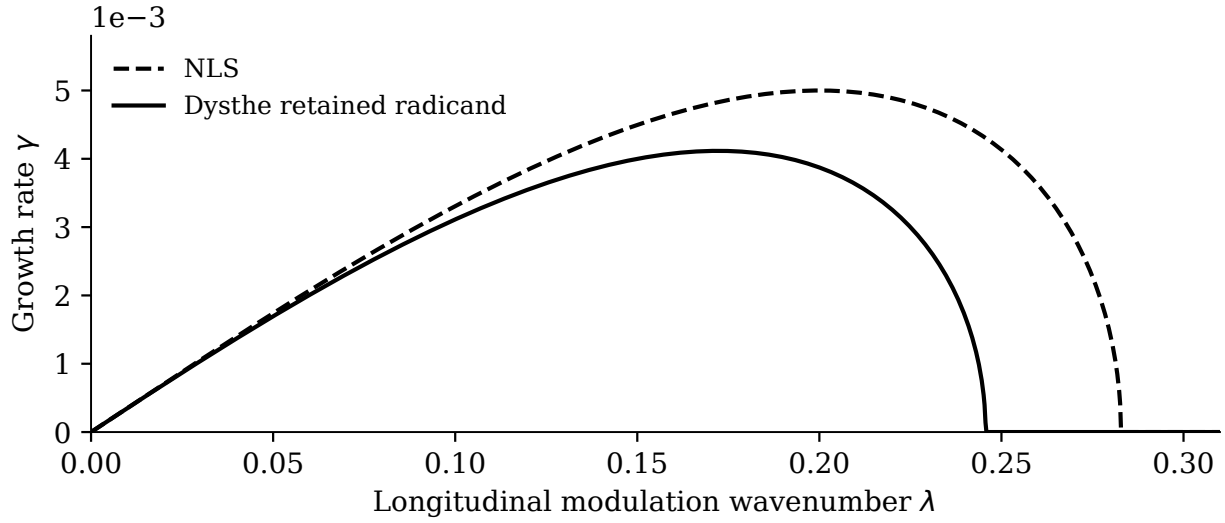


Figure 3: Growth rates from the NLS radicand and Dysthe's consistently truncated radicand, for illustrative $A_0 = 0.10$. These curves compare reduced formulas; no full-equation or experimental data are included.

9.6 Transverse modulations and the paper's figures

For general λ, μ, the retained growing root is

$$\gamma = \sqrt{[A_0^2(1 - \lambda^2/K) - \overline{P}_2]\overline{P}_2}, \tag{9.15}$$

where the radicand must be positive and the wavevector must lie in the slow-modulation regime. For small K, the effective nonlinear factor is positive, so instability requires $0 < \overline{P}_2 < A_0^2(1-\lambda^2/K)$. The current correction depends on the direction of the modulation as well as its size. Dysthe finds the largest retained growth for $\mu = 0$, removing the unbounded maximum-growth hyperbola of NLS.

The paper also discusses a formal large-K cutoff near 3/2, and explicitly notes that this is outside the model's validity range. Do not turn that formal feature into a physical cutoff theorem.

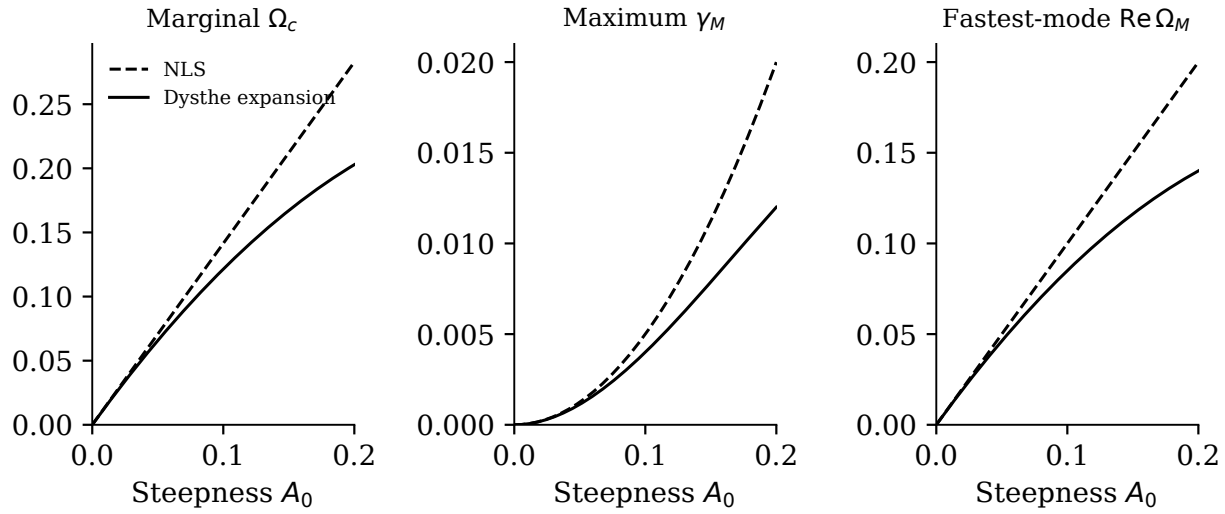


Figure 4: The leading NLS formulas and the small-steepness corrections (3.9), (3.10), and (3.12), plotted only to $A_0 = 0.20$ for illustration. The full-equation and laboratory curves in Dysthe's original figures are not reproduced.

Read the original figures as follows. Figure 1 plots the modulation frequency at marginal stability, not the carrier frequency. Figure 2 plots the maximum exponential growth rate. Figure 3 plots the real frequency separation of the fastest growing sideband, not its growth rate. The paper reports especially good agreement of (3.9)–(3.10) with its reference calculations below steepness roughly 0.3, while (3.12) underestimates the frequency shift above roughly 0.2. These are observable-specific comparisons, not a guarantee of accuracy for every packet throughout that range.

Return to the paper: Read (3.1)–(3.13), printed pp. 110–113. Follow (9.5) to (9.8), then compare each of the three original figure axes with the corresponding formula.

Readiness check. Complete the square in (9.6) and explain why its exact algebraic root is not a higher-order physical result.

10 The long-wave check and a map back to the paper

10.1 Why Whitham theory appears near the end

The paper uses a second approximation method as a consistency check. Whitham modulation theory describes a locally periodic nonlinear wave whose parameters change on scales much longer than its period and wavelength. NLS assumes small amplitude and a narrow band. Whitham's approach need not start with a small-amplitude carrier, but its modulation must be very long. These are overlapping, rather than identical, assumptions.

For a longitudinal perturbation with $\lambda > 0$ tending to zero, (9.8) gives

$$\Omega \simeq \frac{\lambda}{2} + \frac{3}{2}\lambda A_0^2 \pm i\frac{\lambda A_0}{2\sqrt{2}}. \tag{10.1}$$

The λ^3 linear-dispersion correction vanishes in this limit, and $\lambda^2/K = |\lambda|$ tends to zero. The real correction proportional to λA_0^2 survives. This is the stated long-wave result (3.14). For negative λ, the growing branch uses $|\lambda|$ in its positive growth rate.

10.2 Averaged Lagrangian: the minimum needed for (3.15)

A Lagrangian is a function whose variation produces evolution equations. For a nearly periodic wave, average it over the rapid phase to obtain $\mathcal{L}(\omega, k)$ after the carrier-amplitude relation has been incorporated. You do not need a full course in variational mechanics to understand the particular check Dysthe makes.

Let the slowly varying phase be $S(x,t)$, so $k = S_x$ and $\omega = -S_t$. Equality of mixed derivatives implies

$$k_t + \omega_x = 0.$$

Varying the averaged action with respect to S yields

$$\partial_t \mathcal{L}_\omega - \partial_x \mathcal{L}_k = 0.$$

Linearize both equations about constant ω, k. For a modulation $e^{i(\lambda x - \Omega t)}$, the first gives $\delta\omega = c\,\delta k$ with $c = \Omega/\lambda$. Substitute into the second to get

$$c^2\mathcal{L}_{\omega\omega} + 2c\mathcal{L}_{\omega k} + \mathcal{L}_{kk} = 0. \tag{10.2}$$

This is (3.15), in the variables appropriate to that calculation. The second derivatives are entries of the Hessian of $\mathcal{L}$. If $\mathcal{L}_{\omega\omega} \neq 0$, the roots are

$$c = \frac{-\mathcal{L}_{\omega k} \pm \sqrt{\mathcal{L}_{\omega k}^2 - \mathcal{L}_{\omega\omega}\mathcal{L}_{kk}}}{\mathcal{L}_{\omega\omega}}.$$

Negative discriminant means complex modulation speeds and an exponentially growing branch. In PDE language, the real characteristic speeds of the averaged system cease to exist.

The approximate averaged Lagrangian supplied in (3.16) is

$$\mathcal{L} = \frac{\rho g}{2k^2}(\sigma^2 - \sigma^3 - \sigma^4), \qquad \sigma = \frac{\omega^2}{gk} - 1.$$

Here ω denotes the *nonlinear* carrier frequency, with $\omega^2 = gk[1 + (ka)^2 + \cdots]$. Differentiate the function with respect to ω, k before evaluating it on the carrier relation. If you first set $\omega^2 = gk$, you incorrectly set $\sigma = 0$ and erase the nonlinear information the check needs.

Dysthe reports that this calculation agrees with (3.14) at the retained small-amplitude order. That is a cross-check in the common long-wave, small-steepness limit. The polynomial (3.16) is itself an approximation; it is not an exact nonlinear dispersion law. The paper's historical discussion of a long-wave instability boundary near $ka = 0.34$ concerns the separate finite-amplitude Whitham calculation, not a threshold proved by its fourth-order envelope expansion.

10.3 Read the actual paper in three passes

1. **Physical argument.** Read pp. 105–106 and the conclusion on pp. 113–114. Explain the chain: modulation changes averaged momentum flux; a mean current develops; the current shifts phase; the altered phase response changes modulational growth.
2. **Structure of the model.** Read Section 2, identifying the interior equation, boundary conditions, harmonic expansion, scaling, and mean-flow closure. Use the table below. Treat the nonlinear-product algebra as a coefficient-collection calculation rather than memorizing it.
3. **Derivation and evidence.** Work through Section 3 with Section 9 open. Check the determinant, the truncation, and each figure axis. Finish with the long-wave overlap check.

Paper location	What the step does	Background here
Introduction, pp. 105–106	Compares small-amplitude NLS and very-long-wave Whitham regimes; motivates one more asymptotic order.	Sections 1, 5, 6, 10
(2.1)–(2.3), p. 106	States interior and free-surface fluid equations.	Section 3
(2.4)–(2.5), p. 107	Taylor expands surface conditions at $z = 0$.	Section 3
(2.6)–(2.8), p. 107	Separates carrier, harmonics, and mean; finds leading dispersion and group transport.	Sections 2, 4, 5
(2.9)–(2.11), pp. 107–108	Averages to obtain induced mean potential and elevation.	Section 7
(2.12)–(2.13), p. 108	Uses Laplace's equation to eliminate envelope vertical derivatives.	Appendix A.1
(2.14)–(2.16), p. 108	Collects the fundamental harmonic of nonlinear products and its mean-flow interaction.	Sections 2, 5
(2.17)–(2.18), p. 109	Forms an intermediate envelope equation and nondimensionalizes it.	Section 6; A.2
(2.19)–(2.21), p. 109	Reduces time derivatives; states fourth-order equation and mean-flow closure. Check the coefficient issue.	Sections 7–8
(3.1)–(3.5), p. 110	Introduces real amplitude/phase perturbations and linearizes their mean flow.	Sections 6, 9
(3.6)–(3.7), p. 110	Eliminates the mean field and solves a mode determinant to consistent order.	Section 9
(3.8)–(3.12), pp. 111–112	Extracts longitudinal instability band, maximum growth, and frequencies.	Section 9
Figures 1–3, pp. 111–112	Compares different observables with full-equation calculations and some measurements.	Section 9
(3.13), pp. 112–113	Studies oblique modulation and the NLS hyperbola.	Sections 6, 9

Paper location	What the step does	Background here
(3.14)–(3.16), p. 113	Checks the very-long-wave limit using an averaged Lagrangian.	Section 10
(4.1)–(4.3), pp. 113–114	Isolates mean-current feedback as the key growth correction.	Sections 7–8

10.4 What to retain as the paper's conclusion

The paper gives an improved *approximate* description of weakly nonlinear deep-water gravity-wave modulation. Its central stability mechanism is the wave-induced mean current. The asymptotic growth and frequency predictions are supported by the paper's reported comparisons for particular observables, and by an overlap check with long-wave theory.

It does not derive breaking dynamics, prove a uniform error bound for all packets, establish a large-bandwidth model, or validate every later equation called Dysthe. In particular, the original two-horizontal-direction temporal potential-envelope formulation is distinct from a later one-coordinate spatial free-surface-envelope formulation.

Readiness check. Explain why a smaller growth-rate error in Figure 2 does not prove accurate phase prediction or accurate breaking dynamics.

11 Readiness exercises and worked answers

Attempt the exercises before reading the answers. They test the prerequisites and the important transitions in the paper, rather than recall of terminology.

11.1 Exercises

1. For $\zeta = Be^{i\theta} + \text{c.c.}$ with $B = 0.04e^{i\pi/3}\,\text{m}$, write the real surface and its crest amplitude.

2. A carrier has wavelength 8 m and crest amplitude 0.12 m. Calculate steepness and distinguish it from wave height.

3. Starting with $F = z - \zeta(x, y, t)$, derive the kinematic condition.

4. Substitute a deep-water mode into the two linear surface conditions to derive $\omega_0^2 = gk_0$. Explain the choice of vertical exponential.

5. Derive $c_g = c_p/2$, and explain which speed a wave crest follows.

6. If $A = \varepsilon\alpha(\varepsilon(x - t/2), \varepsilon^2 t)$, calculate A_t, A_x, and $A_t + A_x/2$.

7. Find the mean and first-harmonic coefficients of $(Be^{i\theta} + B^*e^{-i\theta})^2$ and of its cube.

8. For $I = I_0 + I_1\cos(\kappa x)$, solve the mean half-space problem and verify its surface current and zero-mode response.

9. In NLS, take $A_0 = 0.10$ and $\mu = 0$. Find the cutoff, maximizing modulation wavenumber, and maximum growth rate.

10. A mode has $\Omega = 0.08 + 0.004i$ in carrier units. Write its time dependence and dimensional growth rate in terms of ω_0.

11. Starting from (9.6), obtain the completed-square root and identify the formally sixth-order contribution.

12. Differentiate the longitudinal Dysthe growth radicand. Show that the maximizing positive wavenumber is $2A_0 - 3A_0^2 + O(A_0^3)$.

13. NLS predicts maximum growth on $\lambda^2 - 2\mu^2 = 4A_0^2$. Why are its arbitrarily large wavevectors not justified predictions?

14. A source changes from temporal potential envelope $A(x, y, t)$ to spatial free-surface envelope $u(\xi, \tau)$. List the conventions that must be checked before comparing coefficients.

11.2 Answers

1. $\zeta = 0.08\cos(\theta + \pi/3)\,\text{m}$. The crest amplitude is 0.08 m, and the leading wave height is 0.16 m.

2. $k_0 = 2\pi/8 \approx 0.7854\,\text{m}^{-1}$, hence $k_0 a \approx 0.0942$. Wave height is approximately 0.24 m; steepness is dimensionless.

3. $DF/Dt = -\zeta_t - \phi_x\zeta_x - \phi_y\zeta_y + \phi_z = 0$. Rearrange to get (3.2), evaluated at $z = \zeta$.

4. Laplace's equation gives vertical factors $e^{\pm k_0 z}$. Only $e^{k_0 z}$ decays into the fluid. The boundary equations are $-i\omega_0 D = k_0 C$ and $-i\omega_0 C = -gD$; combining them yields $\omega_0^2 = gk_0$.

5. $\omega'(k) = \sqrt{g}/(2\sqrt{k})$, whereas $\omega(k)/k = \sqrt{g}/\sqrt{k}$. A crest follows c_p; the narrowband group follows c_g.

6. With $X = \varepsilon(x - t/2)$ and $T = \varepsilon^2 t$, $A_t = -\varepsilon^2\alpha_X/2 + \varepsilon^3\alpha_T$ and $A_x = \varepsilon^2\alpha_X$. Their transport combination is $\varepsilon^3\alpha_T$.

7. The square's mean is $2|B|^2$ and its first harmonic is zero. The cube's mean is zero and its $e^{i\theta}$ coefficient is $3|B|^2B$. The remaining cube terms are at phases $\pm 3\theta$ and the conjugate first harmonic.

8. $\overline{\phi} = -I_1 e^{\kappa z}\sin(\kappa x)$. Then $\overline{\phi}_z = -\kappa I_1 e^{\kappa z}\sin(\kappa x)$ equals I_x at $z = 0$, and $\overline{\phi}_x = -\kappa I_1\cos(\kappa x)$ at the surface. I_0 contributes no forcing or induced derivative.

9. $\lambda_c = 2\sqrt{2}(0.10) \approx 0.2828$, $\lambda_M = 0.20$, and $\gamma_M = 0.005$. These are dimensionless modulation values, not carrier values.

10. The factor is $e^{-0.08it}e^{0.004t}$. Since $t = \omega_0 t_{\text{dim}}$, dimensional growth rate is $0.004\omega_0$ and the amplitude e-folding time is $1/(0.004\omega_0)$.

11. Let $q = \overline{P}_1$ and $b = \lambda A_0^2$. $(q - 5b/4)(q - 7b/4) = (q - 3b/2)^2 - b^2/16$. Moving the last term gives (9.7). $b^2/16$ is sixth order because $b = O(\varepsilon^3)$.

12. Set $s = A_0$. For the nonzero stationary point, $\lambda^2 + 6s^2\lambda - 4s^2 = 0$. The positive root is $-3s^2 + 2s\sqrt{1 + 9s^2/4}$, which expands to $2s - 3s^2 + O(s^3)$.

13. Large $\sqrt{\lambda^2 + \mu^2}$ violates the slow modulation and narrowband assumptions used to truncate the dispersion relation. A polynomial approximation must not be extrapolated outside its scaling.

14. Check the physical envelope, its amplitude normalization, carrier phase convention, evolution coordinate, retarded coordinate, asymptotic ordering, Fourier/Hilbert sign, and which mean-field terms were eliminated. Equality at NLS order does not establish fourth-order equality.

If you can work Exercises 3–12 and explain the answers to 13–14, the main derivation and stability calculation should be accessible. If a specific line remains difficult, refer to its printed equation number; the paper map identifies the relevant prerequisite.

A Two compressed derivative calculations

A.1 Eliminating the carrier envelope's vertical derivatives

Insert $\phi_1 = Ae^{k_0 z}e^{i(k_0 x-\omega_0 t)}$ into Laplace's equation. Differentiate both carrier factors and the slow coefficient. After the k_0^2 contributions cancel, the fundamental coefficient satisfies

$$2k_0 A_z = -2ik_0 A_x - (A_{xx} + A_{yy} + A_{zz}). \tag{A.1}$$

This is (2.12). At leading slow order, $A_z \simeq -iA_x$. Therefore $A_{zz} \simeq -A_{xx}$ and the two longitudinal second derivatives cancel. The next iteration gives $A_z \simeq -iA_x - A_{yy}/(2k_0)$. Because the constant-coefficient derivatives commute, it gives

$$A_{zz} \simeq \left(-i\partial_x - \frac{1}{2k_0}\partial_y^2\right)^2 A = -A_{xx} + \frac{i}{k_0}A_{xyy} + \text{higher slow orders.}$$

Substitution into (A.1) produces (2.13):

$$2k_0 A_z = -2ik_0 A_x - A_{yy} - \frac{i}{k_0}A_{xyy} + O(\varepsilon^5). \tag{A.2}$$

Here A is first order, so the retained A_{xyy} term is fourth order and omitted terms have higher total order. The paper writes its remainder as $o(\varepsilon^4)$. The cancellation is not accidental. A sideband's exact positive vertical exponent is $\sqrt{(k_0+q)^2+r^2}$, which has no purely longitudinal q^2 term near $k_0 > 0$.

A.2 Reducing the second time derivative

In carrier-scaled variables, the left side of (2.17) is

$$2i\mathcal{D}A + \frac{1}{2}A_{yy} - A_{tt} + \frac{i}{2}A_{xyy}.$$

From NLS at third order,

$$A_t = -\frac{1}{2}A_x + F + O(\varepsilon^4), \qquad F = \frac{i}{4}A_{yy} - \frac{i}{8}A_{xx} - \frac{i}{2}|A|^2 A. \tag{A.3}$$

F is third order. At the order needed to differentiate it once, the envelope translates at velocity $1/2$, so $F_t = -F_x/2$ up to fifth-order terms. Thus

$$\begin{aligned} A_{tt} &= -\frac{1}{2}A_{xt} + F_t + O(\varepsilon^5) \\ &= \frac{1}{4}A_{xx} - \frac{1}{2}F_x + F_t + O(\varepsilon^5) \\ &= \frac{1}{4}A_{xx} - \frac{i}{4}A_{xyy} + \frac{i}{8}A_{xxx} + \frac{i}{2}\partial_x(|A|^2 A) + O(\varepsilon^5). \end{aligned}$$

This yields the linear correction $-i(6A_{xyy} - A_{xxx})/8$ on the right of the reduced NLS equation. The derivative of the cubic term also changes nonlinear gradient coefficients.

To expose the source-consistency issue without concealing it, write the dimensionless right-side coefficient of the intermediate equation as

$$\Gamma = |A|^2 + i\beta(AA_x^* - A^*A_x) - i\partial_x|A|^2 + \overline{\phi}_x - i\overline{\phi}_z.$$

The $3ik_0^3$ coefficient printed below (2.17), with the scaling (2.18), gives $\beta = 3/4$. After reducing A_{tt}, the nonlinear gradient terms on the right become

$$i(\beta - \tfrac{1}{2})A(AA_x^* - A^*A_x) - \frac{i}{2}|A|^2A_x. \tag{A.4}$$

For $\beta = 3/4$, the first coefficient is $i/4$. The scan's (2.19) instead prints $3i/2$ and $+i/2$. To obtain the linearized coefficients in (3.2)–(3.3), (A.4) would require $\beta = 2$ and its displayed minus sign. We have not established an authoritative corrected edition or erratum; these are direct algebraic comparisons within the supplied scan.

The practical lesson is to distinguish a formal method you can learn from a printed coefficient chain that may need source correction. The stability determinant derived in Section 9 is independently checked from the stated (3.2)–(3.3). The mean-flow mechanism and NLS prerequisite calculations do not depend on resolving this coefficient issue.

A.3 Verify the discrepancy by linearizing one gradient term

Let the right side of the full NLS balance contain

$$ic\,A(AA_x^* - A^*A_x) + id\,|A|^2A_x + A(\overline{\phi}_x - i\overline{\phi}_z).$$

For $A = A_0e^{-iA_0^2t/2}(1 + \eta u + i\eta v)$, its linear gradient part is proportional to

$$A_0^3\eta\big[id\,u_x + (2c - d)v_x\big].$$

Separating the envelope equation into its real and imaginary parts gives

$$P_1u - \frac{d}{2}A_0^2u_x + P_2v = -\frac{1}{2}\overline{\phi}_z, \qquad P_1v + (c - \tfrac{d}{2})A_0^2v_x - P_2u + A_0^2u = -\frac{1}{2}\overline{\phi}_x.$$

The printed $c = 3/2, d = +1/2$ yield $-1/4, 5/4$ before eliminating $\overline{\phi}_z$. The paper's (3.2)–(3.3) use $+1/4, 7/4$, corresponding to $c = 3/2, d = -1/2$. This supplies a short reproducible check of the sign issue.

B Connection to the project's spatial free-surface model

This appendix identifies a later water-wave convention; it does not replace the original paper's formulation or change the project's approved model.

The 1979 paper uses a temporal envelope, includes two horizontal modulation directions, and introduces A through the velocity potential. A wave-tank experiment may instead prescribe a temporal profile at an upstream location and ask how it changes with propagation distance. This motivates a *spatial* evolution formulation.

Your current project contract identifies a one-coordinate free-surface envelope $u(\xi, \tau)$ with

$$u_\xi = -iu_{\tau\tau} - i|u|^2 u - \varepsilon \left(8|u|^2 u_\tau + 2u^2 u_\tau^* + 2iu|D_\tau|(|u|^2)\right). \tag{B.1}$$

Its contract follows the free-surface equation (1.3) in [3]. Here ξ is propagation distance and τ is retarded time along the profile. In that source's carrier-scaled coordinates,

$$B = \varepsilon u, \qquad \tau = \varepsilon(2x - t), \qquad \xi = \varepsilon^2 x.$$

The companion potential-envelope equation has a different coefficient for the conjugate-gradient term. Equality of leading envelope magnitudes does not justify replacing one by the other at fourth order.

The operator $|D_\tau|$ uses Fourier multiplier $|\kappa|$, with $e^{-i\kappa\tau}$ as the forward transform kernel and zero derivative response for the constant mode. It expresses an eliminated mean-flow response, just as the half-space solution did in Section 7. It is not obtained by simply renaming the x coordinate in (2.19).

Two array axes (ξ, τ) describe an evolution coordinate and one profile coordinate. They do not represent the two horizontal modulation directions (x, y) of the original stability problem. Comparing models requires tracking dependent variable, scaling, direction count, evolution variable, and nonlocal-operator convention.

For learning this paper, retain its notation through the stability derivation. Afterwards, use this appendix to recognize why a later water-wave solver can contain related effects with different-looking coefficients. A numerical implementation should use its separately verified equation contract rather than this pedagogical transcription of the scan.

References

[1] K. B. Dysthe, *Note on a modification to the nonlinear Schrödinger equation for application to deep water waves*, Proceedings of the Royal Society of London A **369** (1979), 105–114. doi:10.1098/rspa.1979.0154. The supplied JSTOR scan is the primary source used throughout this guide.

[2] R. Alan Plumb, *Deep Water Gravity Waves*, Chapter 2 of the notes for MIT 12.333, Atmospheric and Ocean Circulations, Spring 2004. MIT OpenCourseWare chapter and PDF. Optional reinforcement for linear waves and group velocity.

[3] F. Fedele and D. Dutykh, *Hamiltonian description and traveling waves of the spatial Dysthe equations*, arXiv:1110.3605v3 (2012). arXiv source. Used only for the spatial/free-surface comparison in Appendix B; equations (1.2)–(1.3) distinguish the envelopes.